

Tinnaphop Sonsang

111/1 Moo 9, Nong Kratum Sub-district, Thap Than District, Uthai Thani

Email: tinnaphop.moss@gmail.com | Mobile: 064-948-1740

LinkedIn: [Tinnaphop Sonsang](#) | GitHub: github.com/badgaitintinagain, github.com/badgaitintin

A Computer Engineering graduate with hands-on experience building end-to-end machine and deep learning pipelines, from data engineering and model training to real-world deployment. Skilled in computer vision, data engineering, and full-stack web development, with a focus on production-ready systems. Interested in data science, predictive modeling, and applying AI solutions to real-world business problems.

EDUCATION

Walailak University

B.Eng. Computer Engineering and Artificial Intelligence
GPAX: 3.65 (First-class honors) | Graduation: 7 May 2026

Nakhon Si Thammarat

June 2022 – 2026

CERTIFICATIONS

EF SET English Certificate ([View certificate: cert.efset.org/en/gCMcLJ](https://cert.efset.org/en/gCMcLJ))

Awarded 3 Aug 2026

- Overall score: 56/100 — B2 Upper Intermediate (CEFR); Reading: C1 Advanced (64/100)

KEY SKILLS

- Data Gathering & EDA:** Python (Pandas, NumPy), SQL, Matplotlib, Seaborn, Plotly.js | Exploratory Data Analysis (EDA), Data Ingestion & Querying, Anomaly/Outlier Detection, Distribution Analysis & Skewness Correction.
- Data & Feature Engineering:** Feature Transformation, Dimensionality Reduction (PCA, UMAP), Dataset Deduplication via Unsupervised Clustering (KMeans, GMM), STFT Spectrogram Signal Processing, Image Preprocessing & Spatial/Center-Crop Alignment
- Model Training & Evaluation:** PyTorch, TensorFlow, Scikit-learn, OpenCV | Trained and evaluated deep learning models for Image Classification, Object Detection (YOLO), Audio Signal Processing, and Unsupervised Clustering using Precision, Recall, F1-Score, mAP, and SI-SDR.
- Model Deployment & Infrastructure:** Microsoft Azure (Azure ML Compute via SSH Remote Sessions), Cloud GPU/CPU Provisioning, Real-time & Batch Inference Endpoints, RESTful APIs, Hugging Face Spaces, Gradio, Git, Docker & MLOps Fundamentals.
- Web & Data Visualization:** Next.js, React, TypeScript, Astro, Matplotlib, Seaborn, Plotly.js | Exploratory Data Analysis (EDA), Interactive Analytics Dashboards, Full-Stack Integration.

WORK EXPERIENCE

Intern Developer (Co-op/Internship) | ThaiDotRun, Bangkok *Sept 2025 – Apr 2026*

Automated Runner Highlight Extraction System

- Designed an automated video clipping pipeline for running events using an object detection model (RT-DETR) and BoTSORT for subject tracking
- Implemented a custom spatial weighting algorithm to prioritize subjects closest to the camera center, and integrated OCR to extract runner bib numbers, dynamically replacing track IDs to determine precise start and end frames for automated highlight generation
- Achieved an average success rate of approximately 60% for automated highlight video generation, establishing a solid baseline for initial deployment and future optimization

Robustness Shoe Detection & Classification Pipeline

- Designed and developed an end-to-end deep learning pipeline for detecting and classifying brands from complex images and video footage
- Applied unsupervised learning for dataset deduplication to eliminate redundant training frames; used Grounding DINO to detect and localize runners and extract shoe crops for an ensemble learning pipeline trained on the deduplicated dataset
- Implemented depth-mapping and keypoint detection to assess image quality and enhance overall prediction reliability
- Successfully integrated an ensemble classification framework, providing a comprehensive foundation for automated quality control and data re-collection

PROJECTS

Complex-Valued Neural Networks: When Does Phase Actually Help?

July 2026 – Present

- Designed and implemented a complex-valued CNN from scratch (Cartesian weights, tied convolutions, shared-accumulator branching with ModReLU) to systematically benchmark complex representation learning against parameter-matched real-valued baselines on STFT speech classification.
- Formulated and tested 8 hypotheses for the accuracy gap—incorporating Rayleigh weight initialization and complex batch normalization from Trabelsi et al. (2017) alongside capacity-matching and Im/Re weight-drift diagnostics from Mönning & Manandhar (2018)—ruling out capacity mismatch, gradient imbalance, and dead activations.
- Isolated optimization landscape sensitivity as the primary cause of underperformance via multi-seed stability testing, revealing that the complex model exhibits ~8x higher variance across random seeds (~4.9% vs ~0.6% CV) and local-minima sensitivity compared to the real baseline.
- Shifted research focus to phase-critical 2-source audio separation on MiniLibriMix, architecting a complex U-Net utilizing complex ratio masks, SI-SDR loss, and Permutation-Invariant Training (PIT) while strictly maintaining verified parameter equivalence (~1.004x ratio) against real-valued baselines.
- Diagnosed feature attribution using per-layer gradient norms and input gradient heatmaps, resolving skip-connection alignment artifacts via a center-crop architecture fix and isolating current open investigations to time-domain gradient sparsity inherent in SI-SDR and PIT loss formulations.

Madonna Discography Analysis

March 2026 – June 2026

- Performed exploratory data analysis to filter Madonna's tracks out of a multi-artist Spotify dataset and identify anomalies, then developed an end-to-end music analytics pipeline applying unsupervised learning to uncover audio-based patterns across her discography, using a deduplicated set of Spotify audio features
- Addressed heavy right-skew in features such as instrumentality, speechiness, and liveness via Yeo-Johnson power transformation prior to clustering, correcting an initial failure mode where outlier tracks were isolated into non-meaningful micro-clusters
- Benchmarked KMeans against a Gaussian Mixture Model at K=4, selecting the better-fitting algorithm via silhouette score and Davies-Bouldin index, and validated cluster stability using Adjusted Rand Index and bootstrap resampling
- Used UMAP for dimensionality reduction to visualize high-dimensional audio relationships, and auto-generated 4 "Sonic Persona" labels from per-cluster z-scores (e.g., Vocal-driven & Melodic) instead of manual labeling

Sitedotmoss (sitedotmoss.vercel.app)

December 2025 – Present

- Developed a full-stack interactive portfolio deployed on Vercel using Astro (SSR), React 19, TypeScript, and Turso/LibSQL, featuring a tile-based dashboard, a blog with full CRUD operations, and integrated AI inference demos.
- Engineered modular React island components (BlogFeed, InstagramPostModal, BlogCrud) utilizing useState, useMemo, and Set-based data structures for real-time search filtering, tag deduplication, and optimistic UI updates.
- Designed a normalized SQLite schema (5 tables with composite indexes) via Drizzle ORM, building 8 RESTful API endpoints (/api/*) supporting full CRUD, PBKDF2 password hashing, and cookie-based session authentication.
- Integrated real-time AI inference connecting a YOLOv12 white blood cell detection model from Capstone Project on Hugging Face Spaces via Gradio REST API, alongside an interactive Madonna Spotify clustering dashboard powered by Plotly.js.
- Delivered a retro-futuristic Y2K/Frutiger Aero visual system combined with modern minimalism, incorporating glassmorphism UI tokens, dot-matrix canvas animations, and a dynamic glow-tracking Tile component system.

Capstone Project (Thesis) — Detection of White Blood Cells in Raptors

August 2024 – August 2025

- Developed a YOLOv12-based object detection model trained on a custom dataset of 1,200 images to accurately identify six classes of white blood cells in raptors
- Architected a full-stack web application utilizing Next.js, React, and PostgreSQL (via Supabase) to deploy the model, enabling users to upload image batches and receive diagnostic predictions
- Achieved a mean Average Precision (mAP@50) of 0.955, and a mAP@50-95 of 0.742 across multiple IoU thresholds
- Applied Mixup and CutMix data augmentation to improve model generalization, and used Grad-CAM to visualize and validate model attention during evaluation
- Manuscript based on this research is currently in preparation for peer-reviewed publication